

week - 7 : Suppliers Database

Schema Diagram

Suppliers

<u>sid</u>	sname	city	<u>pid</u>	pname	color
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<u>sid</u>	<u>pid</u>	cost
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create table suppliers (sid integer (5)
primary key, sname varchar (20), city
varchar (20));

desc suppliers;

create table parts (pid integer (5) primary key,
pname varchar (20), color varchar (10),
desc parts;

create table catalog (sid integer (5),
pid integer (5), foreign key (sid) reference
suppliers (sid), foreign key (pid) reference
parts (pid), cost float (6), primary key
(sid, pid));

desc catalog;

insert into suppliers values
(10001, 'Acme widget', 'Bangalore');
(10002, 'Johns', 'Kolkata');
(10003, 'Vimal', 'Mumbai');
(10004, 'Reliance', 'Delhi');

select * from ~~parts~~ suppliers;

insert into parts values
(20001, 'Book', 'Red');
(20002, 'Pen', 'Red');
(20003, 'Pencil', 'Green');
(20004, 'Mobile', 'Green');

(20005, 'Charger', 'Black');

select * from parts;

insert into catalog values
(10001, 20001, 10);
(10001, 20002, 10);
(10001, 20003, 30);
(10001, 20004, 10);
(10001, 20005, 10);

(10002, 20001, 10);
(10002, 20002, 20);
(10003, 20003, 30);
(10004, 20003, 40);

select * from catalog;

1 select distinct p-name from parts P,
catalog C where P.ppd = C.ppd;

2 select s-name from suppliers S where
((select count(P.ppd) from parts P) =
(select count(C.ppd) from catalog C
where C.spd = S.spd));

3 select s-sname from suppliers S where
((select count(P.ppd) from parts P where
color = 'Red') = (select count(C.ppd)
from catalog C, parts P where
C.spd = S.spd and C.ppd = P.ppd and
P.color = 'Red'));

4 select P-Pname from parts P, catalog C,
suppliers S where P.ppd = C.ppd and
C.spd = S.spd and S.sname = 'Acme widget'
and not exists (select * from catalog
C1, suppliers S1 where P.ppd = C1.ppd
and C1.spd = S1.spd and S1.sname <>
'Acme widget');

5 select DISTINCT C.spd FROM catalog C
WHERE C.cost > (select max (C1.cost) FROM
catalog C1 where C1.ppd = C.ppd);

6 select P.ppd, S.sname FROM parts P,
suppliers S, catalog C where C.ppd = P.ppd
and C.spd = S.spd and C.cost = (select
max(C1.cost) FROM catalog C1 where
C1.ppd = P.ppd);

OUTPUT:

1) Find the Pnames of parts for which there is some supplier

→ pname:

Book

pen

Pencil

mobile

charger.

2) Find the snames of suppliers who supply every part.

→ Acme widget

3) Find the snames of suppliers who supply every red part.

→ sname

Acme widget

Johns

4) Find the Pnames of parts supplied by Acme widget suppliers and by no one else.

→

Pname

mobile

charger

5) find the sfd of suppliers who charge more for some part than the average cost of that part (averaged over all the suppliers who supply that part.)

→ sfd

10002

10004

6) For each part, find the sname of the supplier who charges the most for that part.

→ pfd

sname

20001

Acme widget

20001

Johns

20002

Johns

20003

Reliance

20004

Acme widget

20005

Acme widget